Triple Contrastive Representation Learning for Hyperspectral Image Classification With Noisy Labels

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***Abstract*— Recently, hyperspectral image classification (HIC) with noisy labels is attracting increasing interest. However, existing methods usually neglect to explore feature-dependent knowledge to reduce label noise and, thus, perform poorly when the noise ratio is high or the clean samples are limited. In this article, a novel triple contrastive representation learning (TCRL) framework is proposed from a deep clustering perspective for robust HIC with noisy labels. The TCRL explores the cluster-, instance-, and structure-level representations of HIC by defining triple learning loss. First, the strong and weak transformations are defined for hyperspectral data augmentation. Then, a simple yet effective lightweight spectral prior attention-based network (SPAN) is presented for spatial–spectral feature extraction of all augmented samples. In addition, cluster- and instance-level contrastive learnings are performed on two projection subspaces for clustering and distinguishing samples, respectively. Meanwhile, structure-level representation learning is employed to maximize the consistency of data after different projections. Taking the feature-dependent information learned by triple representation learning, our proposed end-to-end TCRL can effectively alleviate the overfitting of classifiers to noisy labels. Extensive experiments have been taken on three public datasets with various noise ratios and two types of noise. The results show that the proposed TCRL could provide more robust classification results when training on noisy datasets compared with state-of-the-art methods, especially when clean samples are limited. The code will be available at https://github.com/Zhangxy1999.**

***Index Terms*— Attention mechanism, contrastive learning (CL), deep clustering, hyperspectral image classification (HIC), noisy labels.**

I. Introduction

With the development of remote sensing imaging technology, hyperspectral images (HSIs) have realized the simultaneous acquisition of spectral reflection and spatial information of ground objects. HSI significantly improves the quality of Earth observation and plays an important role in many fields [1], [2], [3]. Therefore, HSI classification (HIC) technology aims to assign a unique label to each pixel and has drawn widespread attention in the community of remote sensing over the last few decades [4]. Although numerous state-of-the-art classification algorithms have been proposed and achieved satisfactory results [5], [6], [7], [8], [9], [10], their success is largely attributed to the supervision information provided by a large number of labels, which are usually believed to be completely accurate [11], [12]. However, in practice, mislabeled samples will inevitably exist, which means that some of the labels used in the training phase are not correctly marked, called noisy labels [13], [14], [15].

Generally, the causes of noisy labels in HSI can be attributed to the following aspects: 1) imprecise positioning in complex surface coverage leads to wrong label assignment; 2) loss of key information during data encoding and transmission; and 3) other mistakes that occur in the manual labeling process [16]. The presence of noisy labels has been a practical issue in remote sensing. In [17] and [18], the experimental results show that noisy labels could mislead the classifier to generate wrong decision boundaries and have a remarkable negative effect on the performance of land-cover classification, which limits the practical application of HSI [12].

For decades, numerous efforts have been made to handle noisy labels in computer vision [19], [20], [21]. Hyperspectral data have high dimensionality, as well as high intraclass spectral variability and interclass spectral similarity, bringing challenges for HIC with noisy labels. Recently, the HIC with noisy labels is also emerging, which can be mainly categorized as follows.

1) *Noisy Label Detection-Based Methods:* For instance, Kang et al. [13] adopted the constrained energy minimization (CEM)-based detection algorithm to detect noisy labels. Tu et al. [22] presented a detection method using the spectral angle mapper of each class and local outlier factor. Also, they developed hierarchical CEM [23] and superpixel-to-pixel weighting distance [24] for the noisy label examples detection. In addition, the density peak clustering algorithm is also introduced to filter out mislabeled samples in [15] and [25].

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2) *Noisy Label Cleaning-Based Algorithms:* Borrowing the idea from semi-supervised learning, Jiang et al. [16] proposed a label noise-cleaning method based on a random label propagation algorithm (RLPA) without the detection stage, which propagates the label information within the superpixel region. Furthermore, the multilayer spectral–spatial graph (MSSG) [14] is presented by performing RLPA via multiscale superpixel segmentation.
3) *Label Noise-Tolerant Loss-Based Methods:* Deep learning plays a promising role in HIC, whereas its performance relies heavily on the quality of labeled samples. In [26], it is indicated that noisy labels can seriously reduce its generalization capability. In very recent years, several works toward robust HIC with noisy labels have been proposed. For example, in [27] and [28], the label noise-tolerant loss, a combination of reverse cross-entropy (CE) loss and normalized CE loss (CELoss), is introduced to HIC with noisy labels, achieving competitive classification performance.

Although these works have achieved robust classification performance under certain conditions, they still have some limitations.

1) *Excessive Data Cleaning:* The labels of clean samples may be corrupted in the mislabeled samples detection and cleaning process, especially when the noise ratio is high or the clean samples are limited.
2) *Known Noise Ratio:* For [14] and [16], noise ratio is always required for the mislabeled sample division; however, the noise ratio is usually unknown in practice. In addition, the above traditional methods need to manually adjust the parameters that have an important impact on the results.
3) *Single Noise Type:* The above works of literature only consider symmetric label noise, and the performance under another more realistic noise type, i.e., asymmetric noise or paired noise, is unknown.
4) *Ignore Feature-Dependent Knowledge:* It is well known that the phenomenon of "heterogenous lander-cover types with similar spectrum" and "the same land-cover with various spectrum" is common in HSI, which means that intrinsic and discriminative features of diverse classes are needed to accurately distinguish clean samples from mislabeled samples. However, the above-mentioned methods ignore to utilize feature-dependent knowledge for HIC with noisy labels, thus performing poorly when the noise ratio is high or the clean samples are limited.

In order to address these issues, this article presents a contrastive learning (CL)-based end-to-end framework. As a self-supervised paradigm, CL [29] has shown powerful representation by learning the intrinsic features of pairwise samples. In [30], CL is integrated into label noise representation learning (LNRL) for exploiting feature-dependent knowledge. The results proved that different views can provide beneficial information for training on the noisy dataset. However, normal CL is usually independent of downstream tasks, which means that the learned features are not directly related to classification. Due to the large intraclass spectral variability and interclass spectral similarity of HSI, discriminative features are needed to identify different classes. Li et al. [31], [32] proposed a clustering task-specified CL via instance- and cluster-level CLs. Inspired by them, we propose an elegant framework called triple contrastive representation learning (TCRL) for robust HIC from the perspective of deep clustering. The proposed TCRL aims to explore cluster-, instance-, and structure-level representations of HIC by defining triple learning loss. Specifically, we first define strong and weak transformations for data augmentation of hyperspectral data. Then, a spectral prior attention-based network (SPAN) is proposed for deep feature extraction of all samples. The cluster- and instance-level CLs are performed on two independent projection subspaces for clustering and distinguishing samples. Meanwhile, structure-level representation learning is employed to maximize the consistency of data after different projections. Instead of detecting mislabeled samples explicitly, the proposed end-to-end TCRL restricts the overfitting of CELoss to noisy labels through triple representation learning to achieve an optimally balanced training state of the model.

Extensive experiments have been conducted on three public datasets under various noise ratios, two noise types, and different network structures to verify the reliability and portability of the proposed framework. The paired noise is considered for the first time in HIC with noisy labels. The experimental results show that the proposed TCRL could achieve more robust performance on noisy datasets compared with state-of-the-art methods.

The contributions of this article are summarized as follows.

1) A TCRL framework is proposed for HIC with noisy labels in an end-to-end manner. To the best of our knowledge, this is the first time that the noisy label problem is considered from a deep clustering perspective. The learned intrinsic feature-dependent knowledge of samples is beneficial to optimize and mitigate the overfitting of CELoss to noisy labels. Meanwhile, the proposed TCRL is a network-independent structure and does not require the noise ratio;
2) An SPAN is proposed for spectral–spatial feature extraction that achieves competitive results with a small number of parameters. In addition, a lightweight spectral prior attention module is proposed to infer the importance of each spectral band for dimensionality reduction of HSI.

The remainder of this article is structured as follows. Section II provides a brief review of some related works. Section III details the proposed framework and network. The experimental results and analysis are presented in Section IV. The conclusion is given in Section V.

II. Related Works

A. Label Noise Representation Learning

Various approaches have been proposed for robust learning with noisy labels [19], [20], [21]. In this section, a brief review of related works on label noise processing based on deep learning is presented. Some of the early approaches work on estimating the noise transition matrix based on the final output

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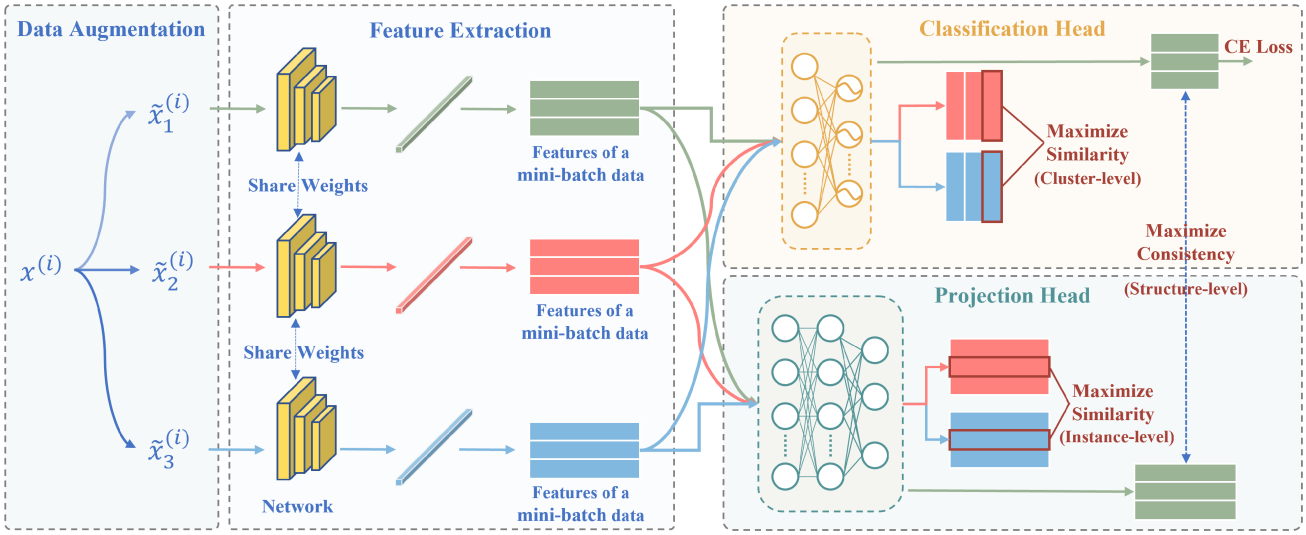


Fig. 1. Flowchart of the proposed TCRL framework. The core ideas of TCRL are that different views can provide beneficial information for training on noisy datasets and TCRL can learn the intrinsic knowledge of samples coming from different classes. Thus, the proposed framework mainly consists of data augmentation, deep feature extraction, and TCRL, i.e., instance-, cluster-, and structure-level representation learnings. This is the first time to consider the problem of noisy labels from the deep clustering perspective, and its contribution is proven in the following ablation experiments.

of the network [33], [34]. However, estimating the noise transfer matrix may be unrealistic as the noise types are complex in reality. From the perspective of the loss function, some studies have tried to develop various robust loss functions for LNRL [35], [36]. Recently, the cotraining strategy is beneficial in learning with noisy labels [37], [38]. For example, coteaching [39] trains two networks simultaneously and updates the networks with each other by selecting small loss data as clean samples during training. Coteaching+ [40] follows a similar scheme as coteaching but updates the model with inconsistent prediction instances. JoCoR [41] also trains two networks, but it promotes their prediction closer together. Unlike them, colearning [30] trains one network but strives to maximize the consistency of data from different views through CL, achieving robust experimental results. Nonetheless, HSIs are quite different from ordinary images, so these methods are not directly applicable to HIC tasks with noisy labels.

In hyperspectral data with high intraclass spectral variability and interclass spectral similarity, intrinsic and discriminative features of different classes are needed to distinguish between mislabeled samples and correct samples. Therefore, different from the aforementioned approaches, we propose to assist LNRL from a new perspective, i.e., deep clustering. The discriminative information exploited by deep clustering is hoped to mitigate the network from overfitting noisy samples. Then, through triple representation learning, an optimally balanced training state of the model can be achieved.

B. HIC Based on Deep Learning

Inspired by the success of deep learning in other fields, many HIC works based on the convolutional neural network (CNN) [8], [42], [43], the recurrent neural network (RNN) [44], the deep belief network (DBN) [45], and Transformer [9], [46] have been studied. Taking advantage of translation equivariance and locality, CNN-based architectures remain dominant in HSI processing [47], [48], [49]. For example, Chen et al. [42] investigated a 3-D CNN-based model for spectral–spatial feature extraction. Zhong et al. [50] proposed a spectral–spatial residual network (SSRN) without feature engineering. In [51] and [52], CNN is combined with a graph convolutional network for HSI structure learning of land covers. To focus on the region of interest in the limited computing resources, the attention mechanism has been increasingly used for feature enhancement, such as the residual spectral–spatial attention network [53] and the double-branch dual-attention network [54]. However, attention mechanisms coming directly from computer vision ignore the prior knowledge specific to the HSI domain and usually require a large computational cost. In this article, a spectral prior-based channel attention module (CAM) is designed for dimension reduction, which effectively suppresses the negative influence of noisy bands.

In addition, although deep learning-based methods have provided satisfactory results in HIC, they usually depend on sufficient carefully labeled data. In practice, however, mislabeled samples are inevitably present in the datasets. Many studies indicate that deep models are susceptible to noisy labels [19]. For robust HIC with noisy labels, a kind of label noise-tolerant loss [35] is introduced to replace the CELoss in [27] and [28]. However, they are just to alleviate the problem of overfitting incorrect labels from the perspective of uncertain labels. The performance under a high noise ratio is not well. Instead, we propose a TCRL framework aiming to exploit feature-dependent knowledge to restrict the overfitting of CELoss to noisy labels through TCRL.

III. Methodology

To achieve robust HIC with noisy labels, this article proposes a framework called TCRL, and the flowchart is shown in Fig. 1. The core idea of the proposed framework can be summarized in the following two aspects.

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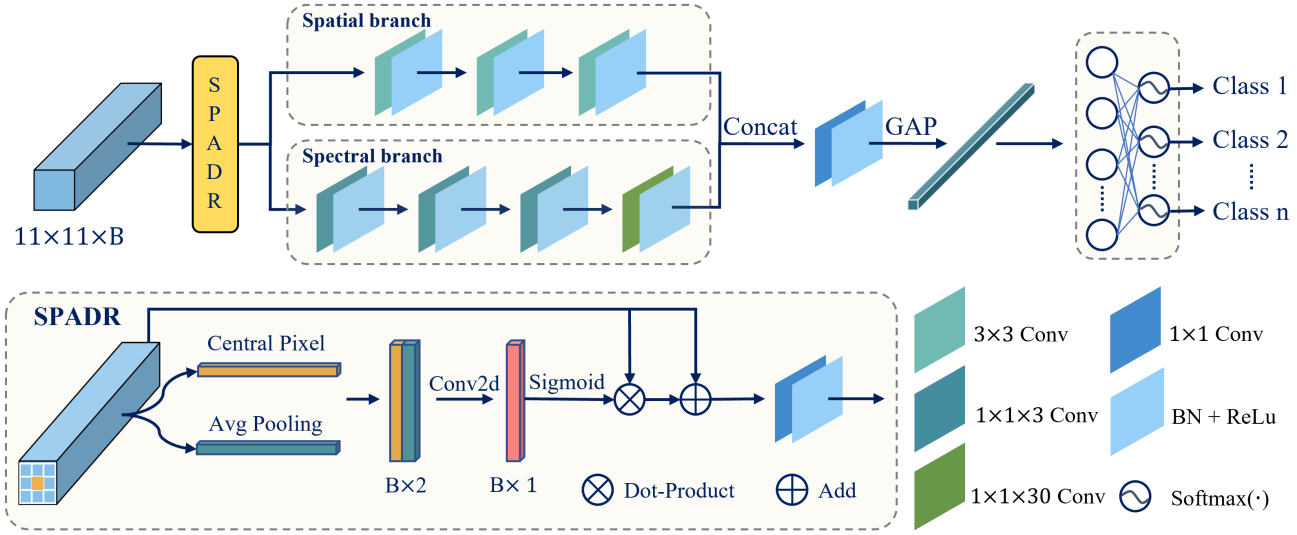


Fig. 2. Flowchart of the proposed SPAN. It mainly consists of three components: SPADR, the spatial feature extraction branch, and the spectral feature extraction branch. The details can be seen in Table I.

1) Different views can provide beneficial information. This motivates us to maximize the similarity of data from different views and the consistency of data after different projections.
2) The cluster-level information learned by deep clustering is helpful to prevent overfitting noisy labels. Therefore, the proposed framework mainly consists of data augmentation, deep feature extraction, and TCRL.

Before that, the preprocessing step is necessary. Given an HSI $I \in \mathbb{R}^{H\times W\times B}$, where H, W, and B represent the height, width, and number of spectral bands of the image, respectively, we first normalize the HSI and then slice it into $H \times W$ patches centered on every pixel, and each patch can be represented by $x \in \mathbb{R}^{L\times L\times B}$, where L is the length of the patch.

A. Data Augmentation

Existing works have shown that the performance of CL relies on appropriate data augmentation [29]. Based on the fact that different views can provide valuable information for training on noisy datasets [30], we have defined two ways of data augmentation for our proposed framework, namely, strong transformation and weak transformation. For the weak transformation, we adopt a simple spatial rotation along the spectral dimension. That is, each patch of the training pixels is randomly rotated by 0°, 90°, 180°, and 270°.

Different from ordinary color images, HSIs have hundreds of channels, so it is difficult to perform data augmentation conventionally. Only the change of spatial dimension is not enough to utilize the characteristics of HSI. Inspired by Song et al. [43], a random spectral shift is employed to interfere with the input patch in the spectral dimension. Random spectral shift refers to the random selection of a part of the spectral bands, where half of the spectral values shift in the horizontal direction and the other half shift in the vertical direction. In this way, we can make good use of the diversity of HSIs. Therefore, the strong transformation is a combination of spatial rotation and random spectral shift. For an input patch $x \in \mathbb{R}^{L\times L\times B}$, we sample one weak transformation to obtain $\{\tilde{x}_1\}$ and two independent strong transformations randomly, and then, the transformed data can be represented as $\{\tilde{x}_k \in \mathbb{R}^{L\times L\times B}\}_{k=1,2,3}$.

B. Network

For efficient HSI feature extraction, a simple yet effective network called SPAN has been developed in this article. The proposed SPAN mainly consists of the spectral prior attention-based dimension reduction (SPADR), the spatial feature extraction branch, and the spectral feature extraction branch. The architecture of the proposed SPAN is shown in Fig. 2.

1) Spectral Prior Attention-Based Dimension Reduction: The attention mechanism usually focuses on the region of interest by calculating the weight of different features. Inspired by CAM in [55], in this article, we propose a lightweight spectral prior-based attention module (SPAM) to acquire the importance of each spectral band for the dimensionality reduction of HSI. The details can be seen in Fig. 2. For an input patch $x \in \mathbb{R}^{L\times L\times B}$, the central pixel is the pixel to be classified, which is the most important clue to infer channelwise attention. To aggregate the spatial information of HSI, the average feature of the patch along the spectral dimension is also taken into account. Therefore, the spectral prior consists of the vector of central pixel $P \in \mathbb{R}^{B\times 1}$ and the average vector $\mu \in \mathbb{R}^{B\times 1}$ of the patch, which considers both spectral and spatial information to facilitate the generation of more accurate attention weights. In addition, inspired by the local cross-channel interaction strategy of efficient channel attention [56], which exploits the channel interaction via 1-D convolution rather than a fully connected (FC) layer with huge parameters, we propose to employ a 2-D convolution layer to obtain the weights. The 2-D convolution layer can also adaptively fuse the features of the average vector and central pixel. The proposed SPAM can be expressed as

$$A = \text{sigmoid}(\text{Conv2D}([P, \mu], W)) \tag{1}$$

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where Conv2D and $W \in \mathbb{R}^{k\times 2}$ denote the 2-D convolution and its weights matrix, respectively. Then, the attention weights $A \in \mathbb{R}^{B\times 1}$ model the importance of each spectral band. In addition, the kernel size k of Conv2D can be adaptively obtained according to the number of spectral bands

$$k = \left| \frac{\log_2 B}{2} + 1 \right|_{\text{ceil}} \tag{2}$$

where $|t|_{\text{ceil}}$ rounds t to the nearest integer larger than it. Then, the output patch after attention weighted can be expressed as

$$X = x \odot A + x \tag{3}$$

in which $\odot$ is the dot product with broadcasting.

After that, the weighted output can be obtained, and different spectral bands have different importance; then, we adopt a convolutional layer with the size of 1×1 for nonlinear dimension reduction. Then, the output of SPADR can be represented as $F \in \mathbb{R}^{L\times L\times \hat{B}}$, where $\hat{B}$ is the reduced number of bands. Following other literature, $\hat{B}$ is set to 30 for all experiments. Through SPAM, the spectral bands of interest can be emphasized, and the negative effects of noisy bands can be suppressed, which is conducive to dimensionality reduction accordingly. Moreover, the proposed SPADR module is beneficial to extract discriminative features and relieve overfitting to noisy labels.

2) Spatial Feature Extraction: A simple spatial feature extraction branch is developed to capture spatial information. In this branch, three convolutional layers with the kernel size of 3×3 are employed, and each convolutional layer is followed by a batch normalization (BN) layer and an rectified linear unit (ReLU) activation function layer. Besides, each convolutional layer has the same number of convolutional kernels without subsampling, which means that the size of the output of the spatial branch is the same as its input.

3) Spectral Feature Extraction: From Fig. 2, the spectral feature extraction branch is similar to the spatial branch. For feature extraction of local spectral sequences, the kernel sizes of the first three 3-D convolutional layers are set to $1 \times 1 \times 3$ and with the same number. Meanwhile, each convolutional layer is followed by a BN layer and an ReLU activation function layer. Furthermore, a 3-D convolutional layer with a kernel size of $1 \times 1 \times 30$ is employed to integrate and learn the whole spectral sequences attribute.

Furthermore, the features learned from the spatial branch and spectral branch are concatenated along the channel dimension. Then, a 2-D convolutional layer with 1×1 kernel size is adopted to fuse the features and reduce data deviation. Finally, the extracted spectral–spatial features after a global average pooling (GAP) layer are fed into an FC layer for classification.

The details of the proposed SPAN are shown in Table I. As can be observed, the proposed SPAN greatly reduces the parameter size through SPADR and small convolutional kernel size. For data input with the size of $11 \times 11 \times 204$, the parameter size of the proposed attention module is only 10. Compared to classic CAM with $204 \times (204/r) \times 2$ parameters (r is the reduction ratio and r is set to 6 in following experiments), our SPAM has negligible parameters. The total number of parameters of the SPAN is about 140k. Some studies show that the overparameterization of deep neural networks is one of the main reasons for overfitting noisy labels [57], [58]. We are hoping that the proposed SPAN with spectral prior and smaller parameters can reduce the degree of overfitting to noisy labels while extracting spectral–spatial features.

TABLE I
DETAILS ON AN EXAMPLE OF THE PROPOSED SPAN

Layer Name		Kernel Size	Output Shape	Param.
Input		-	(204, 11, 11)	-
SPADR	SPAM	(5, 2), 1	(204, 11, 11)	**10**
	Conv2D+BN+ReLU	(1, 1), 30	(30, 11, 11)	6,210
Spatial.	Conv2D+BN+ReLU	(3, 3), 30	(30, 11, 11)	8,190
	Conv2D+BN+ReLU	(3, 3), 30	(30, 11, 11)	8,190
	Conv2D+BN+ReLU	(3, 3), 30	(30, 11, 11)	8,190
Reshape		-	(1, 11, 11, 30)	-
Spectral.	Conv3D+BN+ReLU	(1, 1, 3), 32	(32, 11, 11, 30)	192
	Conv3D+BN+ReLU	(1, 1, 3), 32	(32, 11, 11, 30)	3,168
	Conv3D+BN+ReLU	(1, 1, 3), 32	(32, 11, 11, 30)	3,168
	Conv3D+BN+ReLU	(1, 1, 30), 90	(90, 11, 11, 1)	86,670
Reshape+Concatenate		-	(120, 11, 11)	-
Conv2D+BN+ReLU		(1, 1), 120	(120, 11, 11)	14,760
GAP+Linear		-	(1, 16)	1,936
			Total Param.	**140,684**

C. Triple Contrastive Representation Learning

The proposed TCRL aims to exploit the feature-dependent information for HIC with noisy labels. In TCRL, in addition to CELoss that utilizes the supervised information, the instance- and cluster-level CLs are simultaneously conducted for discriminative representation of different instances and diverse classes, respectively. Furthermore, inspired by Tan et al. [30] and Li et al. [59], a structure-level representation learning is defined to reduce the negative impact of noisy labels on the classifier.

Given a minibatch data $D_t = \{x^{(i)}, y^{(i)}\}_{i=1}^{b}$, in which $x^{(i)}$ and $y^{(i)}$ are the data and label of the ith training sample, respectively, and b is the number of samples in a minibatch. After data augmentation, the transformed data can be expressed as $\{\tilde{x}_k^{(i)}\}_{k=1,2,3}$, and the label is unchanged. Denote $\tilde{x}_1^{(i)}$ as the data after weak transformation, and $\tilde{x}_2^{(i)}$ and $\tilde{x}_3^{(i)}$ are two strongly transformed data. The feature extraction network SPAN with parameters θ_1 takes the transformed patches as inputs and encodes them as the feature vectors $f_k^{(i)} = \text{SPAN}(\tilde{x}_k^{(i)}; \theta_1)$. Note that the features vectors here are the output of the GAP layer of SPAN. Then, these feature vectors are fed into the classification head g with parameters θ_2 and projection head h with parameters θ_3, respectively, to obtain classification probability $\tilde{y}_k^{(i)} = g(f_k^{(i)}; \theta_2)$ and feature projection $z_k^{(i)} = h(f_k^{(i)}; \theta_3)$.

1) Projection Head: In the proposed framework, the instance-level CL is the same as SimCLR [29]. Thus, the nonlinear projection head consists of two FC layers, with

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which the first layer is followed by an ReLU activation function.

The aim of CL is to maximize the similarity of positive pairs while minimizing the similarity of negative pairs. Through instance-level CL, similar samples will be closer, while dissimilar samples will be far apart, which is helpful to alleviate the overfitting of the model to noisy labels. In this article, the positive pairs consist of samples transformed from the same instance, while the negative pairs consist of other instances. For example, given a minibatch of size b and a specific sample $x^{(i)}$, there are $2b-1$ pairs in total (only two strongly transformed data are considered), and a positive pair consists of $\tilde{x}_2^{(i)}$ and $\tilde{x}_3^{(i)}$, while the others pairs are negative pairs. Then, the loss for sample $\tilde{x}_2^{(i)}$ can be defined as

$$l\left(z_2^{(i)}, z_3^{(i)}\right) = -\log \frac{\exp\left(s\left(z_2^{(i)}, z_3^{(i)}\right)/\tau_I\right)}{\sum_{\substack{j=1 \\ j\neq i}}^{b} \exp\left(s\left(z_2^{(i)}, z_2^{(j)}\right)/\tau_I\right) + \sum_{j=1}^{b} \exp\left(s\left(z_2^{(i)}, z_3^{(j)}\right)/\tau_I\right)} \tag{4}$$

where $s(a_1, a_2) = (a_1^T a_2/(||a_1|| \cdot ||a_2||))$ is the pairwise similarity and τ_I is the temperature parameter of instance-level CL. Then, the loss of instance-level CL is in the form of

$$\mathfrak{L}_{\text{Ins}} = \frac{1}{2b} \sum_{i=1}^{b} \left(l\left(z_2^{(i)}, z_3^{(i)}\right) + l\left(z_3^{(i)}, z_2^{(i)}\right)\right). \tag{5}$$

2) Classification Head: The classification head g is the FC layer of SPAN with the Softmax activation function. For supervised multiclass classification, the CELoss is a usually used loss function. In the proposed framework, the CELoss is also used to measure the difference between the classification probability $\tilde{y}_1^{(i)}$ of weakly transformed data and its corresponding label $y^{(i)}$

$$\mathfrak{L}_{\text{ce}}\left(y^{(i)}, \tilde{y}_1^{(i)}\right) = -\sum_{i=1}^{b} y^{(i)} \log\left(\tilde{y}_1^{(i)}\right). \tag{6}$$

Moreover, based on the observation of "label as representation" [31], [32], the cluster-level CL is innovatively introduced for deep self-supervised clustering, which is beneficial to resist the wrong supervision of mislabeled samples. Specifically, the classification probability of each sample in the classification head can be regarded as a special feature representation. In this sense, each column of the probability matrix of a minibatch of data can be regarded as a cluster representation, which should be pushed away from each other. Therefore, similar to instance-level CL, cluster-level CL is carried out on the column space to maximize the cluster representation.

Let $Y_2 = \{\tilde{y}_2^{(i)}\}_{i=1}^{b} \in \mathbb{R}^{b\times C}$ be the probability matrix for a minibatch $\tilde{x}_2^{(i)}$ (and Y_3 for a minibatch $\tilde{x}_3^{(i)}$), in which b equals to the batch size and C is the number of classes. Then, the mth column Y_2^m can be interpreted as the representation of the mth class, and all columns should differ from each other. The similarity between the mth columns of Y_2 and Y_3 should be maximized. Thus, the loss for cluster Y_2^m can be defined as

$$l\left(Y_2^m, Y_3^m\right) = -\log \frac{\exp\left(s\left(Y_2^m, Y_3^m\right)/\tau_c\right)}{\sum_{\substack{j=1 \\ j\neq m}}^{C} \exp\left(s\left(Y_2^m, Y_2^j\right)/\tau_c\right) + \sum_{j=1}^{C} \exp\left(s\left(Y_2^m, Y_3^j\right)/\tau_c\right)} \tag{7}$$

where τ_c is the temperature parameter of cluster-level CL. Then, the loss of cluster-level CL is in the form of

$$\mathfrak{L}_{\text{Clu}} = \frac{1}{2C} \sum_{m=1}^{C} \left(l\left(Y_2^m, Y_3^m\right) + l\left(Y_3^m, Y_2^m\right)\right). \tag{8}$$

Because instance-level CL is usually independent of the downstream classification task, the introduction of cluster-level CL is expected to learn the discriminative representation of different classes for combating noisy labels. In addition, to improve the generalization and slow down the inaccurate convergence of supervised training on noisy datasets, a regularization technique, namely, MixUp [60], which has shown to be effective for learning with noisy labels, is utilized in our framework.

3) Structure-Level Representation Learning: For training on noisy datasets, the above twin CL can simultaneously optimize the feature extraction network; however, the classifier may be misled by noisy labels. Therefore, inspired by Tan et al. [30] and Li et al. [59], a structure-preserving constraint is employed to reduce the negative impact of noisy labels on the classifier, namely, structure-level representation learning.

The basic idea is that the same minibatch of data after different projections should have consistent similarity distribution, that is, a similar relational structure. In this scene, the similarity between $\tilde{y}_1^{(i)}$ and $\tilde{y}_1^{(j)}$ $(i \neq j)$ should be consistent with that between $z_1^{(i)}$ and $z_1^{(j)}$ in the projection head. To reduce deviation, the Euclidean distance $d(\tilde{y}_1^{(i)}, \tilde{y}_1^{(j)})$ is converted to a similarity metric $p(d(\tilde{y}_1^{(i)}, \tilde{y}_1^{(j)})) \in [0, 1]$ by

$$p(d) = e^{-(\sqrt{2}d)^2}. \tag{9}$$

Then, the difference between the similarity distribution of z_1 and $\tilde{y}_1$ can be measured by KL divergence. Thus, the structure-level representation loss is formulated as

$$\mathfrak{L}_{\text{Str}} = \sum_{i\neq j} p\left(d\left(z_1^{(i)}, z_1^{(j)}\right)\right) \log \frac{p\left(d\left(z_1^{(i)}, z_1^{(j)}\right)\right)}{p\left(d\left(\tilde{y}_1^{(i)}, \tilde{y}_1^{(j)}\right)\right)}. \tag{10}$$

4) Objective Function: The optimization of the proposed framework is carried out simultaneously. The overall objective function consists of instance-level contrastive loss, cluster-level contrastive loss, structure-level representation loss, and normal CELoss, i.e.,

$$L = \mathfrak{L}_{\text{ce}} + \mathfrak{L}_{\text{Ins}} + \mathfrak{L}_{\text{Clu}} + \mathfrak{L}_{\text{Str}}. \tag{11}$$

In the training phase, the model is optimized by minimizing (11), and each loss is simply added to achieve a balanced training strategy, while, in the stage of test, only the feature extraction network and classification head are reserved to acquire the classification probability.

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TABLE II
DETAILED INFORMATION OF THREE DATASETS USED IN EXPERIMENTS

UP			Salinas			KSC		
Class	Class Name	Samples	Class	Class Name	Samples	Class	Class Name	Samples
1	Asphalt	6631	1	Brocoli_green_weeds_1	2009	1	Scrub	761
2	Meadows	18649	2	Brocoli_green_weeds_2	3726	2	Willow swamp	243
3	Gravel	2099	3	Fallow	1976	3	CP hammock	256
4	Trees	3064	4	Fallow_rough_plow	1394	4	Slash pine	252
5	Painted metal sheets	1345	5	Fallow_smooth	2678	5	Oak/Broadleaf	161
6	Bare Soil	5029	6	Stubble	3959	6	Hardwood	229
7	Bitumen	1330	7	Celery	3579	7	Swap	105
8	Self-Blocking Bricks	3682	8	Grapes_untrained	11271	8	Graminoid marsh	431
9	Shadows	947	9	Soil_vinyard_develop	6203	9	Spartina marsh	520
			10	Corn_senesced_green_weeds	3278	10	Cattail marsh	404
			11	Lettuce_romaine_4wk	1068	11	Salt marsh	419
			12	Lettuce_romaine_5wk	1927	12	Mud flat	503
			13	Lettuce_romaine_6wk	916	13	Water	927
			14	Lettuce_romaine_7wk	1070			
			15	Vinyard_untrained	7268			
			16	Vinyard_vertical_trellis	1807			
Total		42776	Total		54129	Total		5211

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, extensive experiments, including comparisons with other methods under various noise ratios and ablation experiments, have been conducted on three public datasets to verify the effectiveness and portability of our proposed methodology.

A. Datasets

1) University of Pavia (UP): The first imagery was captured by the ROSIS sensor during a flight in Italy. This dataset has a size of 610 × 340 × 103 with a spatial resolution of 1.3 meters per pixel (mmp). There are 42 776 labeled pixels in total, and they compose of nine land covers.

2) Salinas: The second dataset was acquired by the AVIRIS sensor over the Salinas Valley, CA, USA. This image contains 512 × 217 pixels with a spatial resolution of 3.7 mmp. We use 204 spectral bands in our experiments after 20 water absorption bands were discarded. The Salinas dataset has 54 129 labeled pixels that belong to 16 different classes.

3) Kennedy Space Center (KSC): The third scene was collected by the NASA AVIRIS instrument in 1996. The size of it is 512 × 614 × 176 after discarding water absorption and noisy bands. The ground truth only contains 5211 labeled pixels belonging to 13 categories. The detailed information about three datasets is listed in Table II.

B. Experimental Setup

For comparison, some state-of-the-art approaches for HIC with noisy labels have been considered, including the density peak-based noisy label (DPNL) detection method [25], spatial density peak (SDP) clustering [15], RLPA [16], MSSGs [14], and the dual-channel residual network (DCRN) with noise-robust loss (NRLoss) [27]. Their parameters in experiments are set to default according to their literature, and the classifier of MSSG is the extreme learning machine that achieved the best performance in [14]. Also, the SSRN [50] and the proposed SPAN are validated with noisy labels using CELoss.

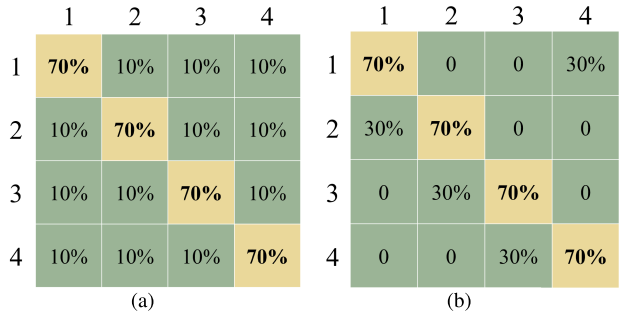


Fig. 3. Noise transition matrix of two kinds of noises in this article. (a) Symmetric noise. (b) Paired noise. $\rho = 30\%$.

In addition, unless otherwise specified, the feature extraction network in the proposed TCRL framework is our proposed SPAN by default. In the TCRL framework, we employ the Adam optimizer, and the learning rate is set to 0.0005. The batch size of TCRL is selected to be 64. For DCRN, SSRN, SPAN, and the proposed TCRL, the training epoch is set to 200 to acquire a stable classification performance.

The noisy labels are generated by manual corruption in this article. It is worth noting that two types of noisy labels, namely, symmetric noise and asymmetric noise, are considered for the first time in HIC with noisy labels. The noise transition matrix of two types of noises in this article is shown in Fig. 3. Let $\rho \in (0, 1)$ denote the noise ratio, and the symmetric noise is uniformly distributed among other classes with probability $(\rho/C - 1)$, which means that the probability of the label reversing to other classes is the same. Different from symmetric noise, asymmetric noise is a more realistic and challenging noise type, which simulates real-world tendencies to confuse some categories during labeling. The asymmetric noisy labels are generated by directly flipping the current class labels to its next class with probability $\rho < 50\%$ in our experiments, for instance, $1 \rightarrow 2, 2 \rightarrow 3, \ldots, 9 \rightarrow 1$ on the UP dataset. Note that the asymmetric noise generated in this article can also be referred to as paired noise. Specifically, we first randomly choose $m + n$ labeled samples from each class and then

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TABLE III

AVERAGE CLASSIFICATION ACCURACY ACHIEVED BY DIFFERENT METHODS ON THE **UP** DATASET UNDER DIFFERENT NOISE TYPES WITH 52 CLEAN SAMPLES AND 24 NOISY SAMPLES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

	52(Clean)+24(Noisy) - Symmetrical noise								52(Clean)+24(Noisy) - Asymmetric noise							
Class	**SPAN**	**SSRN**	**DPNL**	**SDP**	**RLPA**	**MSSG**	**DCRN**	**TCRL**	**SPAN**	**SSRN**	**DPNL**	**SDP**	**RLPA**	**MSSG**	**DCRN**	**TCRL**
1	88.75	89.75	90.21	90.18	86.68	86.78	95.01	**99.56**	94.33	94.35	70.12	86.44	79.47	78.62	95.77	**99.82**
2	96.35	95.79	92.03	92.98	86.34	91.59	98.76	**99.83**	87.60	89.15	92.85	91.66	78.74	87.04	94.46	**99.65**
3	60.96	58.49	51.91	57.10	91.12	91.11	78.74	**91.85**	22.96	27.76	52.97	53.02	**88.29**	87.30	41.90	83.92
4	69.39	75.26	69.47	66.54	73.29	87.34	87.19	**95.37**	77.39	79.50	69.27	61.46	72.77	78.89	87.91	**97.77**
5	61.72	67.77	80.84	82.69	95.53	99.22	93.45	**99.82**	53.22	59.22	63.30	78.28	87.56	96.77	74.14	**98.29**
6	82.33	77.88	50.89	49.26	92.74	96.77	91.37	**95.78**	90.38	86.51	56.84	53.00	83.20	94.69	92.94	**97.09**
7	50.81	50.74	40.06	44.20	97.61	**98.56**	80.80	92.62	43.25	44.71	44.05	43.00	96.96	**98.64**	63.66	94.32
8	70.67	72.72	73.17	77.04	87.79	**91.67**	85.29	90.66	87.80	82.36	81.35	80.69	77.47	90.14	82.50	**94.05**
9	39.46	50.58	83.63	92.88	88.34	95.96	86.34	**98.00**	40.37	45.13	76.30	94.72	77.34	86.82	52.10	**96.39**
OA	79.40	81.50	72.58	73.03	87.23	91.66	92.11	**97.31**	72.40	75.33	71.63	73.76	80.09	86.96	82.68	**97.35**
(std)	(1.92)	(1.87)	(3.00)	(3.05)	(2.39)	(1.48)	(2.41)	**(0.80)**	(1.61)	(2.31)	(2.62)	(2.01)	(3.94)	(1.60)	(4.13)	**(1.34)**
AA	68.94	71.00	70.25	72.54	88.83	93.22	88.55	**95.94**	66.37	67.63	67.45	71.36	82.42	88.77	76.15	**95.70**
(std)	(2.48)	(2.62)	(4.65)	(2.79)	(1.27)	(0.65)	(3.31)	**(1.36)**	(1.12)	(2.00)	(5.10)	(1.57)	(2.95)	(0.70)	(5.12)	**(2.22)**
Kappa	73.76	76.08	65.79	66.47	83.42	89.07	89.70	**96.44**	64.96	68.37	64.37	66.84	74.53	83.07	77.74	**96.51**
(std)	(2.24)	(2.26)	(3.23)	(3.27)	(2.94)	(1.86)	(3.00)	**(1.05)**	(1.90)	(2.71)	(3.01)	(2.07)	(4.74)	(1.94)	(5.03)	**(1.76)**

randomly select n samples from them and corrupt their labels. In this scene, the noise ratio $\rho = (n/(m+n))$. For simplicity, we call the m samples with correct labels clean samples and n samples with wrong labels noisy samples.

All the experiments are repeated ten times with random training samples, and the final results are averaged for presentation. In this article, three evaluating metrics are used for measuring the experimental performance, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa).

C. Comparison With the State-of-the-Art Methods

This section provides a comparison with other advanced approaches in the presence of noisy labels, and the classification accuracy and some full classification maps produced by various methods are also given. On each dataset, we consider two types of noise and different noise ratios.

1) Experiments on the UP Dataset: In this image, following [27], 76 labeled samples per class are randomly selected, among which the labels of 24 samples are randomly corrupted symmetrically and asymmetrically, respectively. The average results achieved by various methods with 52 clean samples and 24 noisy samples are listed in Table III.

When the noise type is symmetric, as can be observed, mislabeled samples detection-based methods, such as DPNL and SDP, achieve poor performance because they may corrupt the clean samples when removing noisy samples. Instead of discarding noisy labels, RLPA and MSSG based on the label propagation algorithm provide competitive classification accuracy. Among them, MSSG employs multiscale superpixel segmentation, so it can achieve more robust results than RLPA that only uses one single-scale superpixel. For deep learning-based methods, both SSRN and the proposed SPAN only adopt CELoss, and they provide lower accuracy than DCRN that utilized the NRLoss function. However, only relying on the NRLoss function cannot prevent the network from overfitting noisy labels. Taking the feature-dependent knowledge learned by triple representation learning, the proposed TCRL framework provides the best results in terms of OA, AA, and Kappa.

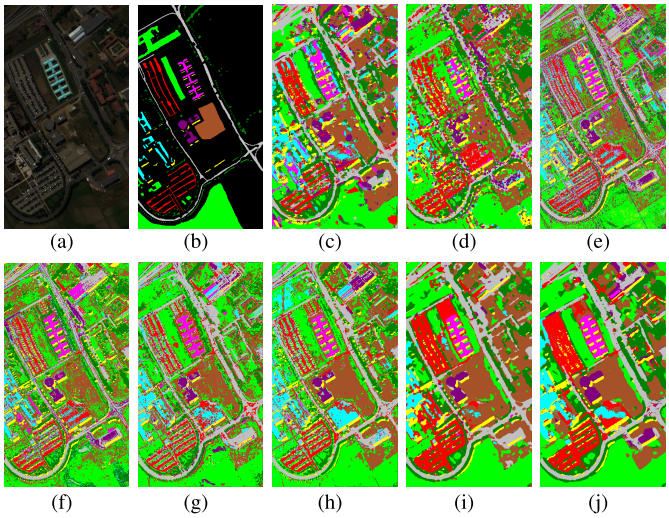


Fig. 4. Classification maps produced by various methods on the UP image with 52 clean samples and 24 mislabeled samples. (a) False-color map. (b) Ground truth. (c) Proposed SPAN. (d) SSRN. (e) DPNL. (f) SDP. (g) RLPA. (h) MSSG. (i) DCRN. (j) Proposed TCRL.

Compared with the standard SPAN only using the CELoss function, TCRL has improved the OA by more than 25% and Kappa by more than 30%, which demonstrates the robustness of the proposed TCRL framework in the presence of noisy labels.

In addition, the full classification maps produced by diverse methods on the UP image are shown in Fig. 4. We can see that SSRN and the proposed SPAN have produced very noisy maps, the reason for that is they overfit the noisy dataset, and the noisy labels mislead the network to produce a wrong decision. Although RLPA and MSSG have achieved high classification accuracy, their classification maps are still noisy because there is no deep extraction of spatial–spectral information for classification. On the contrary, DCRN and our proposed TCRL both obtain a smoother classification map due to the design of techniques robust to noisy labels and the extraction of deep spatial–spectral features. On the one hand, the experimental results show that the proposed TCRL

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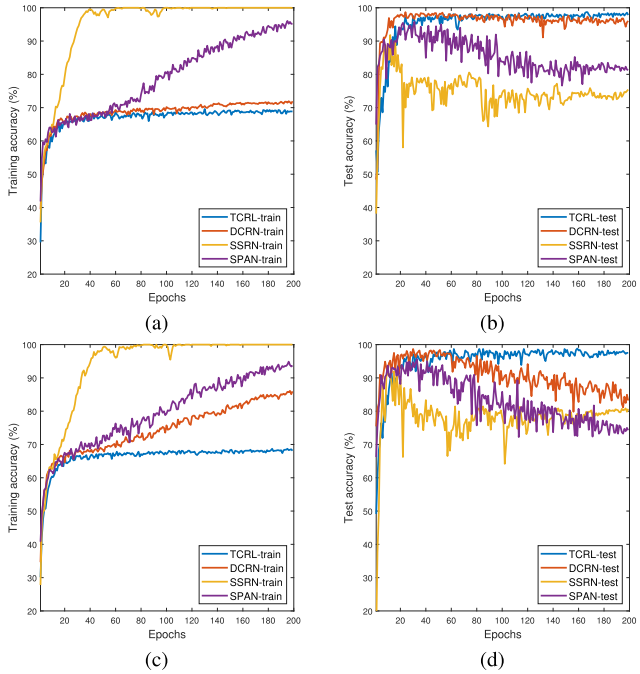


Fig. 5. Curves of training and test accuracy changing with epochs of four deep learning-based methods on the UP dataset under different noise types. (a) and (b) Symmetric noise. (c) and (d) Asymmetric noise. The noise ratio is $\rho = (24/(52+24)) \approx 31.6\%$. We can see that the proposed TCRL does not over fit the noisy labels and achieves a balanced training state.

can effectively mitigate the negative impact of noisy labels. On the other hand, the proposed SPAN can extract deep spectral–spatial features and retain the original structure of the image.

Moreover, the results under asymmetric noise are also listed in Table III. Although the noise ratio is the same as symmetric noise, the accuracy of the comparison methods has decreased to varying degrees. For example, compared with the results under symmetrical noise, the OA of DCRN under asymmetrical noise is reduced by nearly 10%, and that of SPAN decreases by about 7%. This is because there are only two labels for a certain class in the case of asymmetric noise rather than multiple labels in the case of symmetric noise, which means that, under the same noise rate, it is more difficult to determine which label is correct and train robust classifiers in the case of asymmetrical noise. Rather, the proposed TCRL has achieved similar results in the case of asymmetrical noise as in the case of symmetric noise. The reason is that, in addition to the general CELoss, the proposed framework consists of triple representation learning, including instance-level CL for intrinsic feature learning, cluster-level CL for deep clustering and discriminative features learning, and structure-level representation learning for correcting the classifier. This representation learning can learn discriminative features of samples coming from different classes, confining the CELoss overfit noisy labels to achieve a balanced training state. To verify this interpretation, we draw the curves of the training and test accuracy changing with epochs of four deep learning-based methods.

The curves of training and test accuracy changing with epochs are shown in Fig. 5. The noise ratio is $\rho = (24/(52+24)) \approx 31.6\%$. As can be seen, SSRN fits the whole noisy dataset very quickly due to its huge parameters. As the number of training epochs increases, the proposed SPAN was also slowly fitting the noisy data. Therefore, their test accuracy rose first and then declined slowly, as shown in Fig. 5(b). Instead, DCRN and the proposed TCRL did not overfit the noisy labels and, thus, obtained relatively high accuracy under symmetric noise. However, in the case of asymmetric noise, except for the proposed TCRL, the other three methods all overfitted the noisy data and led to low test accuracy. Besides, the training accuracy of the proposed TCRL is not more than 70%, which is consistent with the proportion of clean samples $1 - \rho \approx 68.4\%$. Considering all the results, we can state that the NRLoss function alone cannot effectively prevent the network from overfitting noisy labels in HIC, while the proposed TCRL can mitigate the overfitting of the network to noisy labels with the learned feature-dependent knowledge. Under the constraints of triple learning loss, a balanced training state of the model can be achieved.

TABLE IV

AVERAGE CLASSIFICATION ACCURACY ON THE **UP** DATASET UNDER SYMMETRIC NOISE WITH 20 CLEAN SAMPLES AND 24 NOISY SAMPLES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	SPAN	SSRN	DPNL	SDP	RLPA	MSSG	DCRN	TCRL
1	65.06	56.54	90.35	78.86	56.17	68.82	70.47	**95.06**
2	90.38	84.54	84.09	67.80	51.94	75.02	94.24	**97.38**
3	32.97	24.55	25.17	32.82	73.51	**84.83**	49.51	84.31
4	50.32	50.85	51.31	46.30	47.37	67.23	69.73	**93.56**
5	33.01	31.08	83.02	79.26	72.87	82.00	51.22	**94.22**
6	60.31	45.67	27.28	28.33	77.38	**86.90**	72.36	80.40
7	32.54	21.10	29.88	31.90	93.53	**98.58**	40.12	80.74
8	51.81	42.07	57.68	57.74	63.07	83.60	52.90	**86.02**
9	24.58	24.22	**96.73**	90.14	62.52	81.58	42.23	81.72
OA	58.99	50.58	47.42	48.64	59.39	77.18	67.82	**90.88**
(std)	(3.52)	(3.64)	(6.40)	(8.97)	(3.63)	(3.15)	(4.11)	**(2.00)**
AA	49.00	42.29	60.61	57.02	66.48	80.95	60.31	**88.16**
(std)	(2.37)	(2.08)	(3.62)	(7.39)	(2.68)	(2.08)	(5.39)	**(3.11)**
Kappa	49.66	39.69	40.14	41.07	50.93	71.04	59.95	**88.09**
(std)	(3.77)	(3.68)	(5.87)	(8.02)	(3.80)	(3.70)	(4.57)	**(2.49)**

Furthermore, to verify the performance of the proposed TCRL framework under limited clean samples, we use the same number of noisy samples but only a small number of clean samples for experiments under symmetric noise. On the UP dataset, 20 clean samples and 24 noisy samples per class are used for experiments, meaning the noise ratio $\rho > 50\%$. The average results achieved by different methods are shown in Table IV. It is clear to see that the best results are achieved with our proposed TCRL. However, the other comparison methods all achieve lower accuracy. These experimental results provide strong evidence of the benefits of exploiting feature-dependent knowledge by triple representation learning for noisy label processing.

2) Experiments on the Salinas Dataset: On the Salinas dataset, the training sets consist of 24 clean samples and 12 noisy samples per class. Again, relevant experiments are carried out in the symmetric and asymmetric noise cases, respectively. Table V presents the average results of various methods under different noise types, including accuracy for each class, OA, AA, and Kappa, as well as the standard

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TABLE V

AVERAGE CLASSIFICATION ACCURACY ACHIEVED BY DIFFERENT METHODS ON THE **SALINAS** DATASET UNDER DIFFERENT NOISE TYPES WITH 24 CLEAN SAMPLES AND 12 NOISY SAMPLES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

	24(Clean)+12(Noisy) - Symmetrical noise								24(Clean)+12(Noisy) - Asymmetrical noise							
Class	**SPAN**	**SSRN**	**DPNL**	**SDP**	**RLPA**	**MSSG**	**DCRN**	**TCRL**	**SPAN**	**SSRN**	**DPNL**	**SDP**	**RLPA**	**MSSG**	**DCRN**	**TCRL**
1	88.51	81.26	98.58	23.26	98.54	98.97	99.84	**100.00**	82.63	72.24	87.28	71.96	96.25	**98.79**	98.02	96.02
2	92.48	87.92	98.59	72.46	98.49	99.44	99.58	**99.90**	93.37	83.21	98.44	96.72	90.76	98.69	96.85	**99.49**
3	88.87	75.26	88.54	64.27	**99.79**	99.72	98.17	98.52	82.84	64.07	88.58	87.29	97.08	**99.55**	98.39	98.35
4	87.56	75.94	97.16	79.08	**99.54**	99.37	95.10	94.00	74.99	70.77	97.28	87.46	89.18	**98.59**	97.84	94.26
5	93.94	87.41	95.09	65.49	96.38	96.60	98.26	**99.85**	89.31	84.72	94.50	90.48	96.29	97.36	86.28	**98.12**
6	94.47	90.02	99.31	96.05	97.96	98.88	99.25	**99.93**	86.48	84.60	**99.73**	91.22	92.80	97.17	89.79	96.57
7	89.70	85.57	97.66	84.31	98.65	99.04	99.28	**99.77**	78.16	75.70	96.42	80.95	91.13	**98.19**	97.98	95.56
8	91.09	83.63	68.93	68.33	76.02	78.00	**93.50**	92.99	89.60	82.72	68.64	63.19	68.61	81.11	90.90	**94.03**
9	96.30	93.53	98.67	79.76	98.61	99.36	99.39	**99.41**	68.85	70.75	97.31	81.34	96.27	**98.32**	89.68	95.50
10	89.70	84.84	76.19	75.39	93.99	90.30	95.63	**97.07**	76.11	67.00	84.68	74.18	84.01	88.32	92.32	**96.78**
11	72.27	62.94	90.69	69.67	**98.20**	97.65	91.83	93.39	54.76	48.31	89.46	78.33	**97.05**	96.27	71.11	88.37
12	89.22	79.74	91.90	85.92	99.34	99.49	98.35	**99.99**	83.83	80.75	93.57	94.37	97.88	**99.38**	94.15	98.99
13	78.56	62.15	88.76	93.23	98.10	98.07	90.00	**99.12**	65.44	56.94	88.06	91.40	97.91	**98.15**	79.62	90.92
14	71.54	65.63	93.39	86.25	**97.98**	97.76	86.64	93.30	83.26	74.19	81.00	73.45	96.24	**96.91**	94.53	95.09
15	**82.57**	69.97	51.44	51.16	81.33	80.39	75.69	79.46	81.98	76.51	49.51	47.65	73.96	77.22	80.43	**82.06**
16	83.64	72.47	92.12	84.51	97.09	97.54	97.80	**99.75**	44.03	39.19	96.76	66.50	88.78	**98.16**	80.71	88.15
OA	88.35	80.5	82.75	72.39	91.10	91.44	92.69	**94.58**	77.93	72.43	81.22	73.89	85.44	91.17	88.50	**93.16**
(std)	(1.04)	(1.41)	(1.93)	(5.73)	(0.70)	(0.86)	(2.63)	**(0.86)**	(1.73)	(2.17)	(2.49)	(8.53)	(1.64)	(1.22)	(4.10)	**(1.77)**
AA	86.9	78.64	89.19	73.7	95.62	95.66	94.89	**96.64**	77.23	70.73	88.2	79.78	90.89	**95.14**	89.91	94.27
(std)	(1.45)	(1.38)	(1.76)	(8.25)	(0.35)	(0.49)	(1.64)	**(0.62)**	(1.69)	(1.48)	(1.21)	(6.72)	(2.22)	**(0.76)**	(4.60)	(1.06)
Kappa	87.08	78.42	80.88	69.45	90.11	90.48	91.90	**93.96**	75.64	69.55	79.17	71.06	83.86	90.18	87.24	**92.40**
(std)	(1.15)	(1.54)	(2.08)	(6.23)	(0.78)	(0.95)	(2.88)	**(0.96)**	(1.89)	(2.41)	(2.39)	(9.54)	(1.82)	(1.34)	(4.52)	**(1.97)**

deviation. As can be seen, most approaches have provided high accuracy under symmetric noise, including the standard SPAN with the CELoss function. However, in the case of asymmetric noise, the accuracy of SPAN has a great reduction. On the contrary, the proposed TCRL framework has achieved similar results in both symmetric and asymmetric noise cases, indicating that the TCRL framework is robust and effective for noisy labels. Besides, a similar accuracy has been achieved by MSSG under two cases on the Salinas dataset. The reason is that MSSG is a superpixel-based method, and on the Salinas dataset, multiscale superpixel segmentation can carve out the spatial information of each class accurately, facilitating the cleaning of noisy labels. On images with more complex image textures, such as UP and KSC, however, MSSG does not perform so well. The proposed TCRL framework can achieve good generalization on multiple datasets.

Fig. 6 illustrates the classification maps on the Salinas dataset with 24 clean and 12 noisy samples under symmetric noise. Both SPAN and SSRN using only CELoss have achieved inaccurate classification maps. Compared with SPAN, the proposed TCRL framework using SPAN as a feature extractor provides a smoother classification map, proving its effectiveness in coping with noisy labels.

Moreover, the experiments with limited clean samples are also conducted on the Salinas dataset, and the average results are listed in Table VI. Note that there are only eight clean samples, and 12 noisy samples per class are randomly selected for training. Since our proposed TCRL explores discriminative information of samples coming from different classes, as can be seen, it outperforms DCRN by close to 10% in terms of AA.

3) Experiments on the KSC Dataset: On the KSC image, the training sets are composed of 24 clean samples and 12 noisy samples per class. The averaged experimental results under two noise types are reported in Table VII. A similar

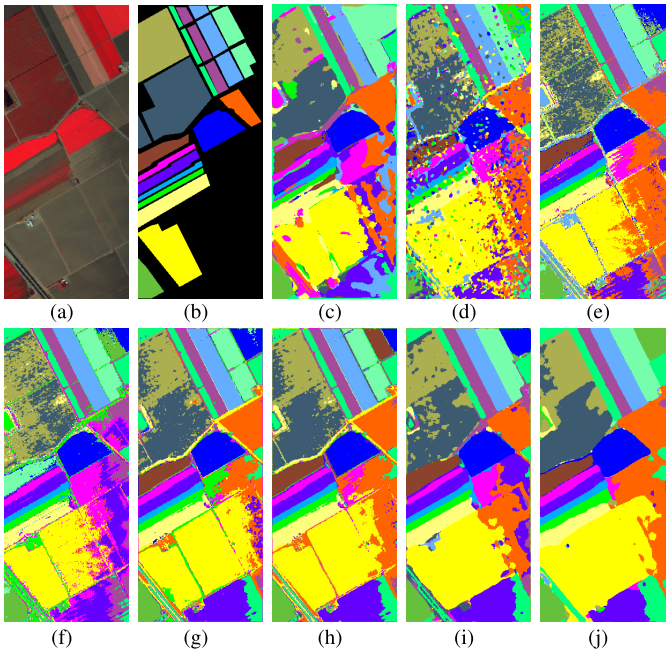


Fig. 6. Classification maps produced by various methods on the Salinas image with 24 clean samples and 12 noisy samples. (a) False-color map. (b) Ground truth. (c) Proposed SPAN. (d) SSRN. (e) DPNL. (f) SDP. (g) RLPA. (h) MSSG. (i) DCRN. (j) Proposed TCRL.

conclusion can be drawn from these results. The proposed TCRL framework has achieved the best result in terms of OA, AA, and Kappa. Also, some classification maps with 24 clean samples and 12 noisy samples under symmetric noise are shown in Fig. 7. The proposed TCRL still produces a smoother map compared with other methods. In addition, the performance with limited clean samples under symmetric noise is also tested, and only eight clean samples and 12 noisy samples per class are randomly chosen for training. The results

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TABLE VI
AVERAGE CLASSIFICATION ACCURACY ACHIEVED BY DIFFERENT METHODS ON THE **SALINAS** DATASET UNDER SYMMETRIC NOISE WITH EIGHT CLEAN SAMPLES AND 12 NOISY SAMPLES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	SPAN	SSRN	DPNL	SDP	RLPA	MSSG	DCRN	TCRL
1	66.69	52.68	94.79	28.87	97.36	**97.92**	81.93	97.91
2	70.53	69.19	94.82	32.66	84.99	**99.36**	92.46	98.94
3	42.04	46.83	76.83	51.45	93.58	**96.47**	80.57	94.10
4	46.00	37.49	86.36	55.38	85.27	**97.39**	82.92	92.22
5	68.63	63.29	83.29	67.81	96.09	94.73	92.61	**99.15**
6	68.45	70.84	92.47	84.16	67.21	92.06	88.44	**98.65**
7	64.57	60.48	89.22	45.91	90.13	94.10	91.80	**99.28**
8	75.70	66.24	60.69	56.38	54.61	68.14	81.59	**82.17**
9	83.86	81.75	92.96	73.80	86.45	98.02	98.54	**98.69**
10	57.37	49.73	72.43	64.69	78.36	78.32	91.74	**97.50**
11	40.79	32.64	73.90	37.11	**98.41**	94.42	71.20	87.88
12	49.83	48.19	84.47	75.90	98.62	**99.07**	90.00	98.03
13	23.46	22.83	77.90	80.99	**98.52**	98.19	64.92	95.16
14	34.88	33.61	73.93	66.80	93.88	**95.38**	73.83	85.34
15	63.24	53.06	43.88	42.75	58.65	**73.83**	63.44	63.49
16	52.08	42.56	85.74	69.30	84.61	96.96	85.03	**99.87**
OA	59.35	54.06	73.89	59.01	76.35	86.36	80.71	**88.25**
(std)	(3.72)	(2.59)	(3.12)	(9.16)	(2.98)	(2.22)	(4.00)	**(3.18)**
AA	56.76	51.96	80.23	58.37	85.42	92.15	83.19	**93.02**
(std)	(3.12)	(2.43)	(3.92)	(11.95)	(1.62)	(0.90)	(3.71)	**(1.80)**
Kappa	55.54	49.79	71.34	54.98	74.02	84.89	78.68	**86.95**
(std)	(3.98)	(2.76)	(3.33)	(9.71)	(3.21)	(2.43)	(4.40)	**(3.51)**

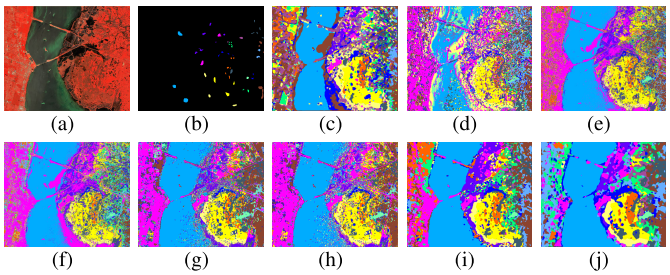


Fig. 7. Classification maps produced by various methods on the KSC image with 24 clean samples and 12 noisy samples. (a) False-color map. (b) Ground truth. (c) Proposed SPAN. (d) SSRN. (e) DPNL. (f) SDP. (g) RLPA. (h) MSSG. (i) DCRN. (j) Proposed TCRL.

are listed in Table VIII, and it is obvious that the proposed TCRL still performs the best.

D. Effectiveness of the Proposed Attention Module

To clarify the contribution of the proposed SPADR, we conduct relevant experiments using different dimension reduction methods, such as principal component analysis (PCA), nonlinear 1×1 convolution (1×1 Conv), and SPADR. The results are listed in Table IX. It can be observed that the proposed SPADR has achieved the best results. Compared with only using 1×1 Conv, adding SPAM before dimension reduction results in a positive effect. This indicates that the proposed spectral prior-based attention can effectively enhance the valuable bands by utilizing the prior knowledge of HIC and is conducive to dimension reduction.

Furthermore, we compare the proposed SPAM with the classic CAM in [55]. It employs two FC layers to compute attention with the max-pooling and average pooling of the data. We simply replace our SPAM with CAM to weigh each spectral band and then use 1×1 convolution for dimension reduction. We can call it CAM-based dimension reduction (CADR). It is clear that the CAM has a large negative impact on the results. The reason may be that the CAM designed without using domain knowledge in HIC cannot effectively deliberate important information of the raw HSI data. In addition, the large number of parameters in CAM may be more easily misled by noisy labels. In contrast, our SPAM uses the spectral prior that directly indicates the features of interest and only has few parameters, thus improving the performance.

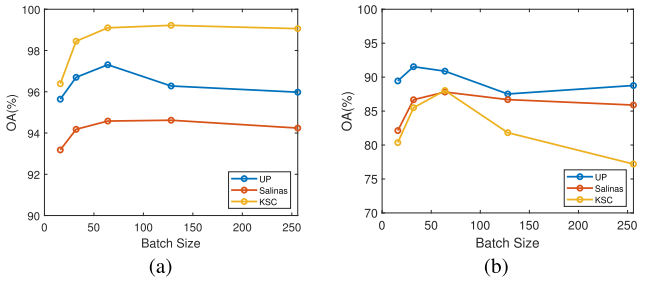


Fig. 8. Effect of different batch sizes on the performance of the proposed TCRL. (a) 52 clean and 24 noisy samples are selected for the UP dataset; 24 clean and 12 noisy samples are chosen for the Salinas and KSC datasets. (b) 20 clean and 24 noisy samples are selected for the UP dataset; eight clean and 12 noisy samples are chosen for the Salinas and KSC datasets. The batch sizes are 16, 32, 64, 128, and 256, respectively.

E. Ablation Experiments

In this section, we have conducted extensive ablation experiments on three datasets under two different noise ratios. We investigate the influence of different batch sizes on classification performance. In addition, the importance of each component is also studied.

1) Effect of Batch Sizes: The batch size is set to 16, 32, 64, 128, and 256, respectively. Fig. 8 presents the experimental results. Usually, in CL, a larger batch size can have enough negative sample pairs to achieve better results. However, this conclusion may not apply to HIC. We can see that, as the batch size increases, the accuracy first increases and then decreases on all three datasets. Since the number of training samples in HIC is not as large as in the ImageNet dataset, the batch size cannot be too large. As we can see, setting a larger batch size results in a sharp decrease in accuracy when clean samples are limited. Therefore, the batch size in TCRL is set to 64, which is an appropriate batch size for the data size.

2) Contribution of Each Representation Learning: The role of each representation learning in the proposed TCRL framework is validated. The experimental results are presented in Table X. Compared with the standard SPAN that only uses the CELoss, TCRL has made a remarkable improvement when training on the noisy dataset. As can be seen, each component contributes to the overall results. Compared to instance-level CL, however, cluster-level CL and structure-level representation learning have a larger impact on final results. In TCRL, instance-level CL and cluster-level CL are hoping to learn the discriminative representation of different instances and diverse classes, respectively. Therefore, cluster-level CL is more significant for the classification task. Despite that the

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TABLE VII

AVERAGE CLASSIFICATION ACCURACY ACHIEVED BY DIFFERENT METHODS ON THE **KSC** DATASET UNDER DIFFERENT NOISE TYPES WITH 24 CLEAN SAMPLES AND 12 NOISY SAMPLES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

	24(Clean)+12(Noisy) - Symmetrical noise								24(Clean)+12(Noisy) - Asymmetrical noise							
Class	**SPAN**	**SSRN**	**DPNL**	**SDP**	**RLPA**	**MSSG**	**DCRN**	**TCRL**	**SPAN**	**SSRN**	**DPNL**	**SDP**	**RLPA**	**MSSG**	**DCRN**	**TCRL**
1	92.50	87.25	94.26	96.78	97.93	97.70	98.95	**99.95**	86.99	67.59	93.38	90.22	97.39	97.24	92.39	**100.00**
2	75.06	59.33	81.67	76.58	51.69	50.53	94.73	**97.99**	42.78	42.28	77.64	80.26	52.17	45.27	**93.10**	86.59
3	81.00	69.08	81.45	84.43	97.18	96.18	97.56	**99.84**	72.77	67.54	73.34	74.15	72.86	82.05	**96.92**	93.77
4	76.18	65.11	62.02	57.43	35.69	33.29	93.99	**97.70**	70.75	62.81	58.78	55.99	32.92	1.53	75.09	**90.64**
5	66.77	47.16	58.79	58.19	73.12	58.72	82.87	**95.47**	59.24	53.57	46.18	46.57	**83.04**	82.80	66.24	71.38
6	75.27	62.73	59.49	50.22	48.45	32.18	**95.72**	95.08	80.25	74.24	63.05	55.36	46.06	37.67	**95.34**	94.19
7	62.94	39.25	67.29	63.09	**100.00**	**100.00**	93.03	89.86	51.67	42.63	57.49	48.75	**100.00**	**100.00**	88.52	93.00
8	87.84	74.46	75.60	67.28	95.80	91.82	97.57	**99.58**	94.26	90.80	57.87	60.98	86.35	84.15	99.14	**99.83**
9	90.78	76.79	86.38	84.67	87.38	83.74	97.90	**99.91**	79.08	73.56	77.27	76.10	91.49	63.26	85.32	**96.78**
10	84.73	71.71	93.16	91.26	89.89	91.14	97.30	**100.00**	67.62	66.34	78.67	72.87	89.29	88.94	91.12	**91.65**
11	90.67	76.05	83.89	94.83	95.95	97.81	98.72	**99.18**	71.76	70.15	80.69	82.65	93.47	**96.45**	89.27	91.78
12	89.23	79.39	90.46	84.49	70.94	64.26	99.03	**99.45**	78.98	73.52	81.38	90.51	64.30	57.28	**98.49**	96.11
13	96.21	90.83	96.88	99.63	96.22	96.21	99.26	**100.00**	85.39	82.83	90.47	93.34	94.84	**96.03**	84.31	95.68
OA	86.69	74.58	84.72	83.36	85.39	82.99	97.30	**99.05**	75.98	69.84	77.08	77.41	82.71	77.72	89.34	**94.01**
(std)	(1.65)	(2.22)	(1.22)	(1.72)	(1.47)	(2.34)	(0.95)	**(0.59)**	(2.63)	(1.48)	(7.03)	(5.32)	(1.89)	(3.55)	(4.33)	**(3.05)**
AA	82.24	69.17	79.33	77.61	80.02	76.43	95.89	**98.00**	72.42	66.76	72.02	71.37	77.25	71.74	88.87	**92.41**
(std)	(1.56)	(2.05)	(0.97)	(1.47)	(1.88)	(3.41)	(1.49)	**(1.37)**	(2.40)	(1.71)	(6.28)	(4.96)	(2.15)	(3.52)	(2.70)	**(3.03)**
Kappa	85.12	71.74	82.96	81.47	83.65	80.93	96.97	**98.94**	73.19	66.36	74.41	74.76	80.64	75.02	88.08	**93.27**
(std)	(1.85)	(2.41)	(1.35)	(1.91)	(1.63)	(2.62)	(1.07)	**(0.66)**	(2.91)	(1.64)	(7.78)	(5.91)	(2.11)	(3.99)	(4.81)	**(3.40)**

TABLE VIII

AVERAGE CLASSIFICATION ACCURACY ACHIEVED BY DIFFERENT METHODS ON THE **KSC** DATASET UNDER SYMMETRIC NOISE WITH EIGHT CLEAN SAMPLES AND 12 NOISY SAMPLES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	**SPAN**	**SSRN**	**DPNL**	**SDP**	**RLPA**	**MSSG**	**DCRN**	**TCRL**
1	84.06	73.82	85.88	85.24	88.19	86.59	94.30	**96.11**
2	66.32	45.20	66.08	60.32	39.60	29.28	70.79	**87.42**
3	62.72	42.04	64.45	59.49	**85.42**	80.04	72.40	81.11
4	63.34	37.86	25.87	37.47	7.03	13.41	**63.67**	60.21
5	46.37	27.91	42.20	44.52	**81.35**	65.60	49.89	55.71
6	59.65	44.08	38.66	34.91	45.84	50.00	71.40	**86.45**
7	53.17	30.76	56.69	47.35	99.53	**99.65**	69.28	71.47
8	77.59	52.76	48.07	50.96	77.18	47.32	80.84	**88.67**
9	86.09	60.16	70.92	63.16	56.84	55.96	86.65	**97.01**
10	67.34	47.09	63.29	65.35	75.89	67.84	73.78	**99.33**
11	81.60	65.10	88.54	78.35	**92.03**	91.18	89.89	90.69
12	73.71	53.99	69.68	66.43	64.51	51.10	83.28	**94.61**
13	91.39	84.19	93.21	92.77	93.78	92.93	94.15	**99.65**
OA	71.94	54.92	67.12	63.11	74.27	68.61	79.29	**87.76**
(std)	(5.78)	(3.50)	(6.65)	(9.96)	(5.09)	(8.12)	(4.86)	**(4.91)**
AA	70.26	51.15	62.58	60.49	69.78	63.92	76.95	**85.26**
(std)	(5.44)	(3.13)	(5.02)	(7.95)	(3.88)	(6.59)	(5.29)	**(6.34)**
Kappa	68.92	50.16	63.65	59.38	71.38	65.04	76.97	**86.43**
(std)	(6.28)	(3.64)	(7.08)	(10.66)	(5.56)	(8.84)	(5.34)	**(5.24)**

TABLE IX

CLASSIFICATION RESULTS PRODUCED BY THE PROPOSED SPAN USING DIFFERENT DIMENSIONALITY REDUCTION METHODS

Dataset		PCA	1×1 Conv	SPADR	CADR
UP	OA	77.61±1.80	77.81±1.81	**79.40±1.92**	73.17±1.98
	AA	66.72±2.57	68.25±2.36	**68.94±2.48**	61.83±2.16
	Kappa	71.54±2.05	71.84±2.12	**73.76±2.24**	66.16±2.19
Salinas	OA	79.34±1.37	87.82±0.83	**88.35±1.04**	75.98±1.55
	AA	74.83±1.23	86.62±1.17	**86.90±1.44**	72.39±2.11
	Kappa	77.17±1.53	86.47±0.92	**87.08±1.15**	73.51±1.70
KSC	OA	84.03±2.05	83.73±1.94	**86.69±1.65**	74.82±2.25
	AA	78.78±1.89	78.54±1.81	**82.24±1.56**	69.40±2.12
	Kappa	82.19±2.21	81.93±2.17	**85.12±1.85**	72.01±2.48

results' improvement brought by instance-level CL is relatively small, we would like to emphasize that the final robust results are achieved by joint representation. Although the twin CL can jointly optimize the feature encoder, the classifier head may also be inevitably disturbed by noisy labels. The structure-level representation learning used to correct the classifier from a data consistency perspective also plays an important role.

In addition, we can see that, with a small number of clean samples, the absence of cluster-level contrastive loss leads to a significant reduction in both accuracy and stability, which is a good demonstration of the irreplaceable position of discriminative cluster representation learned by deep clustering in TCRL. It also proves that, on hyperspectral data with high interclass spectral similarity and intraclass spectral variability, dealing with noisy label problems from a deep clustering perspective is a proper attempt. This is consistent with our motivation.

F. Performance With Different Numbers of Noisy Samples

Furthermore, to evaluate the performance of different methods, we have conducted comparison experiments under different noise ratios. Specifically, with reference to literature [27], we fixed the number of clean samples and achieve different noise ratios by varying the number of noisy samples. On the UP dataset, the number of clean samples for each class is 50. Then, the number of noisy samples per class is 5, 25, 50, 75, and 100. The OA, AA, and Kappa for the diverse methods at different noise ratios are shown in Fig. 9(a)–(c). For both the Salinas and KSC datasets, 24 clean samples are randomly selected, and the number of noisy samples for each class is 6, 12, 24, 36, and 48. The maximum noise ratio is approximately 66.67% on all three datasets.

As can be seen in Fig. 9, the accuracies of all methods decrease to varying degrees as the noise ratio increases.

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TABLE X
ABLATION EXPERIMENTS ON THREE DATASETS WITH TWO NOISE RATIOS UNDER SYMMETRIC NOISE

Dataset	Method	OA	AA	Kappa	OA	AA	Kappa
UP	Samples	52 (Clean) + 24 (Noisy) samples			20 (Clean) + 24 (Noisy) samples		
	TCRL	**97.31±0.80**	**95.94±1.36**	**96.44±1.05**	**90.88±2.00**	**88.16±3.11**	**88.09±2.49**
	w/o $\mathfrak{L}_{Ins}$	95.93±1.69	92.97±3.01	94.66±2.18	89.96±2.05	86.38±2.60	86.91±2.53
	w/o $\mathfrak{L}_{Clu}$	95.24±0.79	91.54±1.60	93.76±1.02	71.42±12.97	76.05±9.48	64.35±14.73
	w/o $\mathfrak{L}_{Str}$	92.79±2.11	87.39±3.87	90.60±2.66	84.44±3.36	79.16±3.61	79.91±4.11
Salinas	Samples	24 (Clean) + 12 (Noisy) samples			8 (Clean) + 12 (Noisy) samples		
	TCRL	**94.58±0.86**	**96.65±0.62**	**93.96±0.96**	**88.25±3.18**	**93.02±1.80**	**86.95±3.51**
	w/o $\mathfrak{L}_{Ins}$	94.18±0.88	96.23±0.91	93.53±1.05	87.11±3.78	91.92±2.40	85.71±4.14
	w/o $\mathfrak{L}_{Clu}$	93.10±0.69	95.18±1.09	93.22±0.77	79.28±4.52	84.32±6.04	77.09±4.90
	w/o $\mathfrak{L}_{Str}$	92.36±1.17	93.06±1.32	91.51±1.32	82.49±3.45	84.80±3.95	80.60±3.80
KSC	Samples	24 (Clean) + 12 (Noisy) samples			8 (Clean) + 12 (Noisy) samples		
	TCRL	**99.05±0.59**	**98.00±1.37**	**98.94±0.66**	**87.76±4.91**	**85.26±6.34**	**86.43±5.24**
	w/o $\mathfrak{L}_{Ins}$	98.33±0.59	97.58±1.25	98.13±0.67	86.99±6.65	85.24±7.50	85.59±7.32
	w/o $\mathfrak{L}_{Clu}$	97.37±0.70	96.26±1.34	97.17±0.78	67.34±9.25	72.88±6.57	63.74±9.93
	w/o $\mathfrak{L}_{Str}$	94.87±0.89	92.64±1.25	94.26±1.00	80.54±9.19	79.31±8.66	78.49±9.10

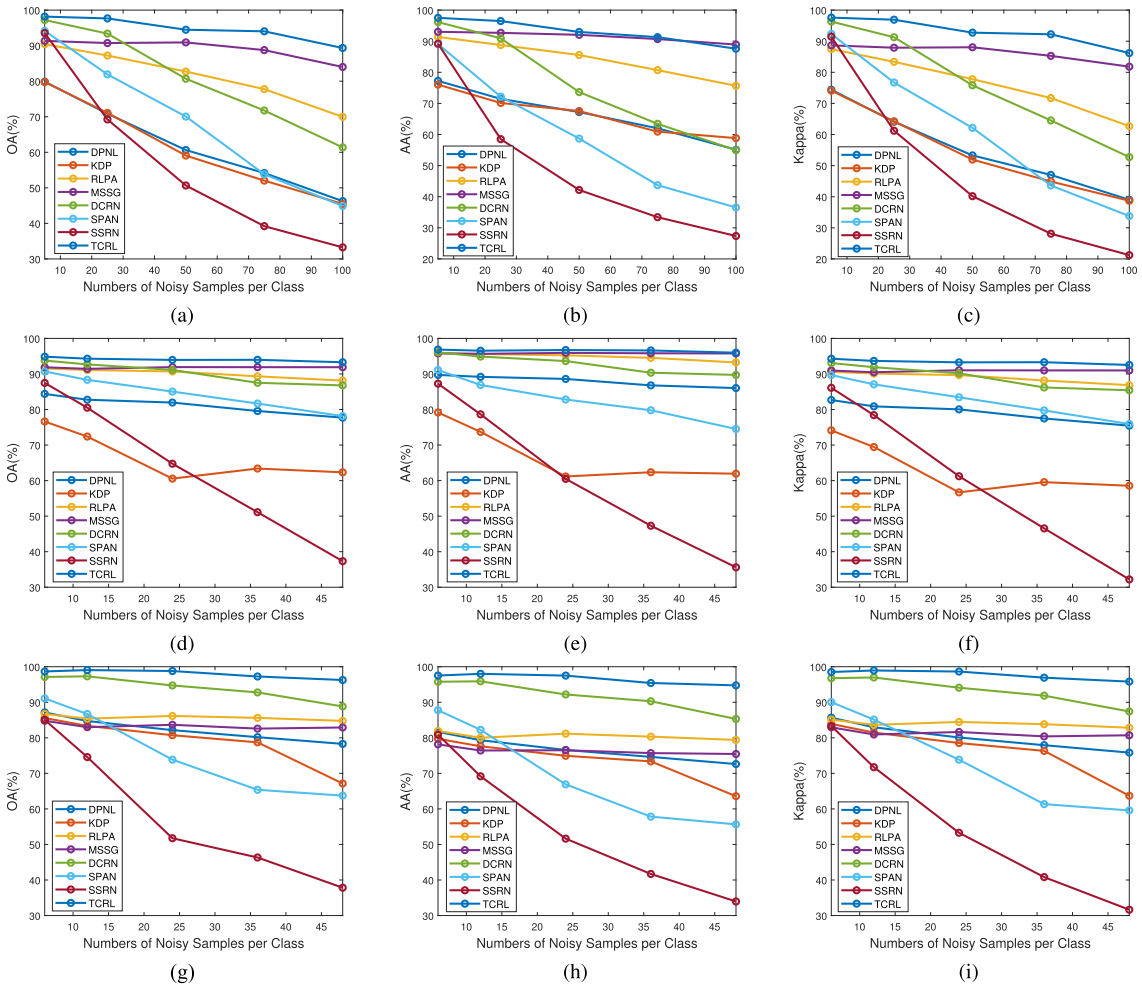


Fig. 9. Performance of various methods with different noisy samples under symmetric noise on (a)–(c) UP dataset, (d)–(f) Salinas dataset, and (g)–(i) KSC dataset.

However, our TCRL framework consistently maintains a high accuracy rate. Even when the noise rate is at its maximum, the proposed TCRL still achieves an OA of roughly 90% on the UP dataset, whereas DCRN that employs the NRLoss function only achieved an accuracy of about 60%. This shows that exploring feature-dependent information is significant for robust HIC with noisy labels. Furthermore, compared with SPAN, SSRN, and DCRN, the accuracy of TCRL is the

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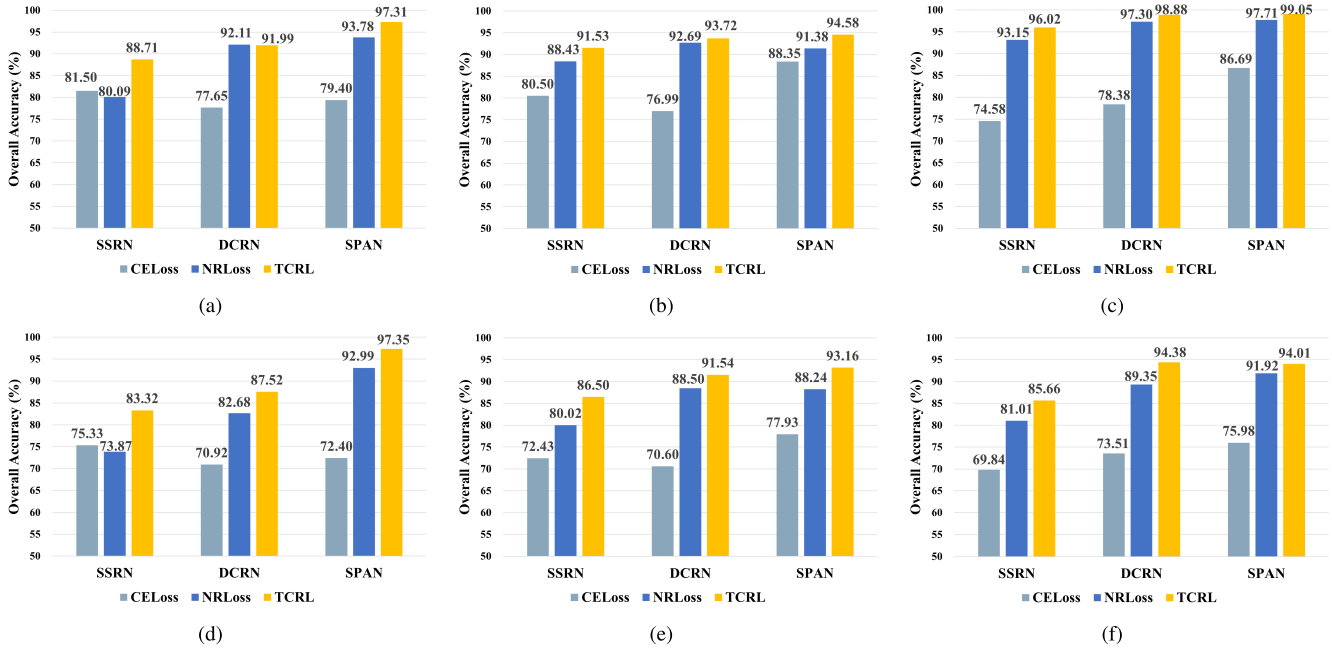


Fig. 10. Portability analysis of the proposed TCRL using different networks and the comparison between different training strategies on the UP, Salinas, and KSC datasets under two noise types. (a)–(c) Symmetric noise. (d)–(f) Asymmetric noise.

least degraded. These results demonstrate that TCRL can remain effective in the case of high noise ratios through triple representation learning.

Considering all the above results, it is possible to state that, taking the feature-dependent information exploited by TCRL, the proposed TCRL framework is effective and robust for HIC with noisy labels.

G. Portability Analysis

To validate the portability of the proposed TCRL framework, we have conducted extensive experiments based on three different networks (i.e., SSRN, DCRN, and SPAN) under symmetric and asymmetric noises. If the input size is uniformly $11 \times 11 \times 204$, the parameter sizes of SSRN, DCRN, and SPAN are 370k, 331k, and 140k, respectively. In addition, each network is optimized using three separate training strategies, i.e., normal training only using CELoss, only NRLoss function, and our proposed TCRL. The OA obtained on three datasets is shown in Fig. 10.

The following information can be obtained from the results. On the one hand, with the same training strategy, our proposed SPAN achieves competitive results on all three datasets compared to the other two networks, demonstrating that the SPAN can effectively extract spatial–spectral features despite its simple design. On the other hand, apart from the abnormal results of SSRN on the UP dataset, the training accuracy generated by the same network structure can be significantly improved when the training strategy is NRLoss or TCRL compared to CELoss. In addition, the proposed TCRL can provide a more robust and stable performance than NRLoss, especially in the case of asymmetric noise [see Fig. 10 (second row)]. These results fully illustrate the effectiveness of our proposed TCRL on the noisy label classification problem and its portability. Meanwhile, it is also proved that exploring the feature-dependent knowledge is significant for HIC with noisy labels.

V. Conclusion

In this article, a TCRL framework aiming to exploit feature-dependent information has been proposed for robust HIC with noisy labels. Under the constraints of triple learning loss, i.e., cluster-level contrastive loss, instance-level contrastive loss, and structural-level learning loss, an optimally balanced training state can be achieved on noisy datasets. Extensive experimental results conducted on three datasets have demonstrated the effectiveness and portability of the proposed TCRL framework. It is worth noting that this is the first time to consider the problem of noisy labels from the deep clustering perspective. The results demonstrate that cluster-level representation can play a significant role when training on noisy datasets. Meanwhile, it can be stated that exploring feature-dependent knowledge is conducive to preventing overfitting noisy labels. In addition, the proposed spectral prior attention module for hyperspectral data has also shown excellent performance.

Furthermore, we introduce asymmetric noise into the HIC with noisy labels for the first time. The experimental results show that, even if the noise ratio is the same as that of symmetric noise, the accuracy of existing methods in the case of asymmetric noise decreases to varying degrees. Compared with other methods, the proposed TCRL shows higher accuracy and better robustness on datasets with noisy labels. While, in this article, we consider the noisy label problem under class-balanced conditions, however, in remote sensing data, the distribution of ground objects is usually very unbalanced. In the future, we will focus on the problem of HIC with noisy labels under the long-tailed distribution.

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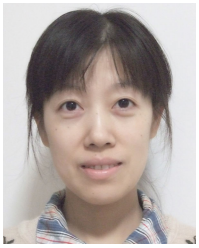
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